Jorge Sánchez Canales

PhD candidate in Governance, Hertie School

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Education

PhD — **Hertie School, Centre for Sustainability**, Governance — Berlin, Germany — Feb 2024 – present

- Supervisor: Prof. Lion Hirth.
- Research on electricity markets: market power, demand response and subsidy design.
- Contributing to the projects Ariadne II and Machine Learning in Electricity Market Research (ML-Strom).

Master of Public Policy — **Hertie School**, Public Policy (concentration in Policy Analysis) — Berlin, Germany — 2021 – 2023

- Average grade: 1.3 (German scale, where 1.0 is the best).
- Specialization in sustainability and energy economics, with courses on Economic Growth and Climate Change, Net-Zero Energy Systems, Energy Economics, Electricity System Modelling and Electricity Market Design.
- Master thesis: Energy crises, price caps and side effects. A causal inference study of the Iberian cap on gas using OLS.

Bachelor — **Universidad Carlos III de Madrid, Universitat Pompeu Fabra and Universidad Autónoma de Madrid**, Philosophy, Politics and Economics — Madrid, Spain — 2016 – 2020

- Average grade: 9.02/10 (Spanish scale).
- Beca de Excelencia, Comunidad de Madrid (2019).
- With distinction in Research Methodologies for Political Science and Economic Sociology, among others.

Experience

Barcelona School of Economics, Visiting PhD student — Barcelona, Spain — Sept 2026 – Nov 2026 — 3 months

Host: Prof. Mar Reguant.

Hertie School, Centre for Sustainability, Research Assistant and Research Associate — Berlin, Germany — Feb 2023 – Feb 2024 — 1 year 1 month

Supported the research of several PhD candidates in the field of electricity markets.

- Discussed methodology and study design.
- Collected and transformed data from different databases.
- Designed and implemented statistical analyses in Python 3.
- Produced result visualizations and supported the writing of academic manuscripts.

Mercator Research Institute on Global Commons and Climate Change, Research Assistant — Berlin, Germany — Sept 2022 – July 2023 — 11 months

Meta-analyses of the empirical literature on electricity markets and behavioural interventions.

- Performed systematic searches and selection of literature.
- Coded, extracted and standardized data from the relevant literature.
- Supported the synthesis of results, including analysis and visualization, and the writing of academic manuscripts.

Teaching

Economic Growth and Climate Change: Teaching assistant, Master's level, Hertie School. Spring 2024 (45 teaching hours). Instructor: Prof. Lion Hirth.

Energy Economics 1: Teaching assistant, Master's level, Hertie School. Spring 2025, Spring 2026 and Fall 2026 (12 teaching hours per term). Instructor: Prof. Lion Hirth.

Energy Economics 2 (formerly Electricity Market Design): Teaching assistant, Master's level, Hertie School. Spring 2025 (as Electricity Market Design), Spring 2026 and Fall 2026 (12 teaching hours per term). Instructor: Prof. Lion Hirth.

Master's thesis advising: Co-advised six Master's theses at the Hertie School (three in 2024-25 and three in 2025-26), with Prof. Lion Hirth as official supervisor.

Publications

An empirical estimate of the electricity supply curve from market outcomes Sept 2026
Jorge Sánchez Canales, Alice Lixuan Xu, Chiara Fusar Bassini, Lynn H. Kaack, Lion Hirth
10.1016/j.eneco.2026.109523 (Energy Economics, 161, 109523)

Revealing the empirical flexibility of gas units through deep clustering Sept 2025
Chiara Fusar Bassini, Alice Lixuan Xu, **Jorge Sánchez Canales**, Lion Hirth, Lynn H. Kaack
arxiv.org/abs/2504.16943 (ECML PKDD 2025 Proceedings, Springer Lecture Notes in Computer Science)

Behavioral, Information, and Monetary Interventions to Reduce Energy Consumption in Households: A Living Systematic Review and Network Meta-Analysis Dec 2025
Tarun M. Khanna, Diana Danilenko, Qianyi Wang, Luke A. Smith, Bhumika T. V., Aditya Narayan Rai, **Jorge Sánchez Canales**, Tim Repke, Max Callaghan, Mark Andor, Julian H. Elliott, Jan C. Minx
10.1002/cl2.70070 (Campbell Systematic Reviews, 21(4), cl2.70070)

Revise and Resubmit

Identifying elasticities in autocorrelated time series using causal mechanism graphs 2026
Lion Hirth, Jonas Peters, Oliver Ruhnau, **Jorge Sánchez Canales**, Felix Schur, Raffaele Sgarlato, Silvana Tiedemann
arxiv.org/abs/2409.15530 (Revise and resubmit at Econometrica)

Market power abuse in wholesale electricity markets June 2025
Alice Lixuan Xu, **Jorge Sánchez Canales**, Chiara Fusar Bassini, Lynn H. Kaack, Lion Hirth
arxiv.org/abs/2506.03808 (Revise and resubmit at Energy Economics)

Working Papers

De-risking renewable energy investments: Assessing contract design and project finance using operational wind park data May 2026
Jorge Sánchez Canales, Lion Hirth
arxiv.org/abs/2605.23400 (Working paper)

Work in Progress

Demand response with behind-the-meter assets: what can we learn?
Draft in progress.

Skills

Programming: Python 3 (advanced), Pyomo (intermediate)

Office: Microsoft Word, Excel and PowerPoint

Languages

Spanish: Native

English: Excellent (C1)